

LABORATORY RECORD

PYTHON AUTOMATION

Selenium Automation - Module 2 Technical Report

Submitted by

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TECHNICAL REPORT

MODULE 2 — UNIT TEST FRAMEWORKS & PAGE OBJECT MODEL

Scope

Three completed Selenium/Python practical assignments covering login automation, data-driven testing with external test data, and PyTest integration with automated HTML reporting.

No.	Assignment
7	Login Automation Script
8	Data-Driven Testing (DDT) using External CSV
9	PyTest Integration with HTML Reporting

All scripts target the SauceDemo (<https://www.saucedemo.com>) practice application. Execution screenshots are reproduced from the original terminal and browser sessions.

ASSIGNMENT 7

Login Automation Script

Description

Automate the login flow on the SauceDemo application: launch the browser, maximize the window, navigate to the login page, submit valid credentials, and validate that the resulting URL matches the expected post-login inventory page.

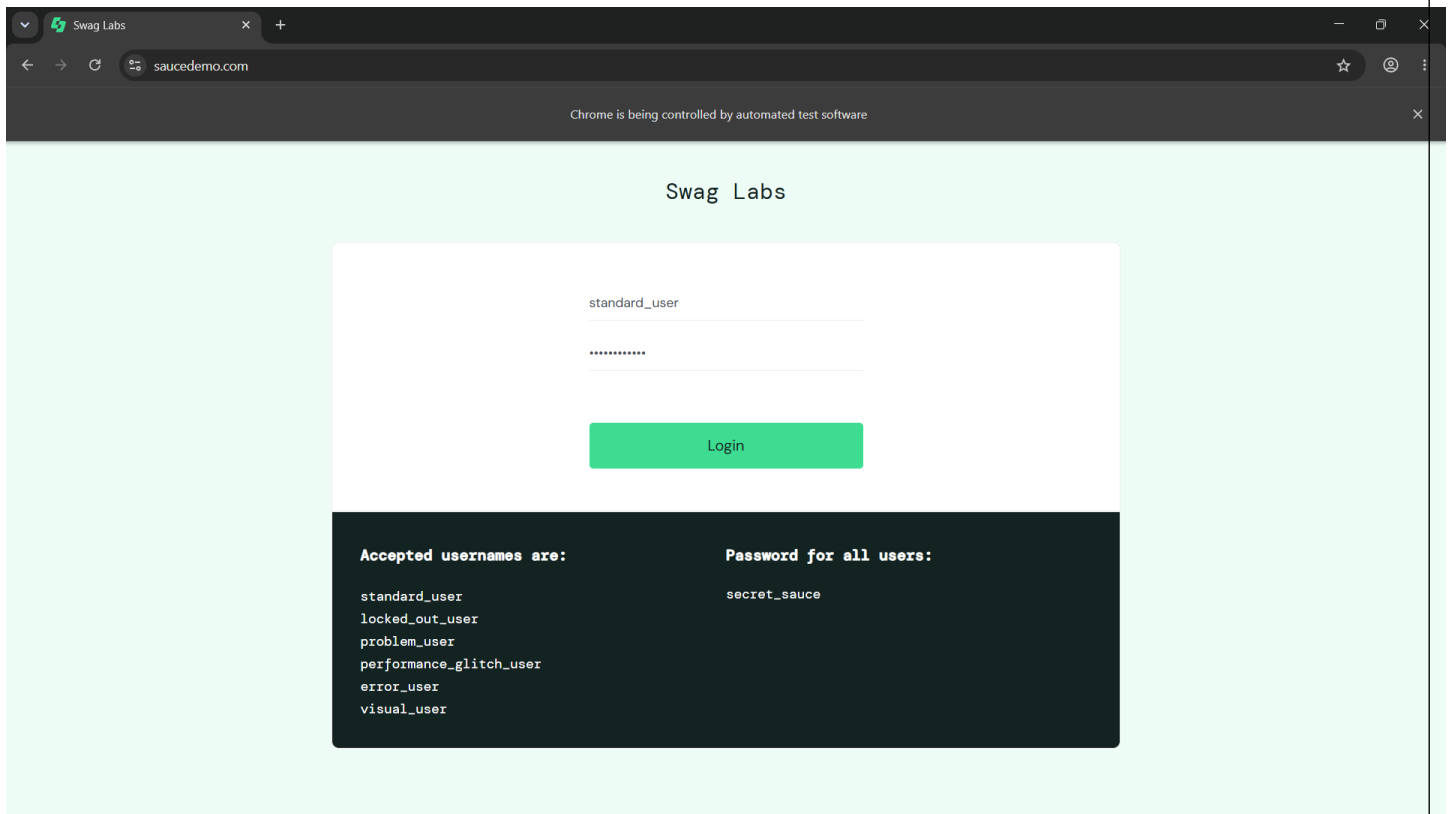
Python Selenium Implementation

```
Q7.py X
Module 2 > Q7.py > ...
1
2 from selenium import webdriver
3 from selenium.webdriver.common.by import By
4 import time
5
6 driver = webdriver.Chrome()
7 driver.maximize_window()
8 driver.get("https://www.saucedemo.com/")
9 driver.find_element(By.ID, "user-name").send_keys("standard_user")
10 driver.find_element(By.ID, "password").send_keys("secret_sauce")
11 time.sleep(1)
12 driver.find_element(By.ID, "login-button").click()
13 time.sleep(2)
14 assert driver.current_url == "https://www.saucedemo.com/inventory.html"
15 print("ASSIGNMENT 7 - LOGIN TEST PASSED")
16 print("Current URL:", driver.current_url)
17 print("Page Title:", driver.title)
18 driver.quit()
```

Execution Evidence

```
Problems Output Debug Console Terminal Ports
• (venv) PS D:\PythonAutomation\Assignments\Module 2> python Q7.py
ASSIGNMENT 7 - LOGIN TEST PASSED
Current URL: https://www.saucedemo.com/inventory.html
Page Title: Swag Labs
(venv) PS D:\PythonAutomation\Assignments\Module 2> |
```

Terminal output confirming successful login and correct redirect URL.



Browser session showing credentials entered on the SauceDemo login page.

ASSIGNMENT 8

Data-Driven Testing (DDT) using External CSV

Description

Build a login script that reads multiple test cases from an external CSV file instead of hardcoding credentials. Loop through each row, attempt login with the given username/password combination, capture the resulting validation error message, and assert it matches the expected error defined for that row.

Test Data (test_data.csv)

```
test_data.csv X
Module 2 > test_data.csv > data
1  username,password,expected_error
2  locked_out_user,secret_sauce,"Sorry, this user has been locked out"
3  standard_user,wrong_password,do not match any user
4  ,secret_sauce,Username is required
5  standard_user,,Password is required
```

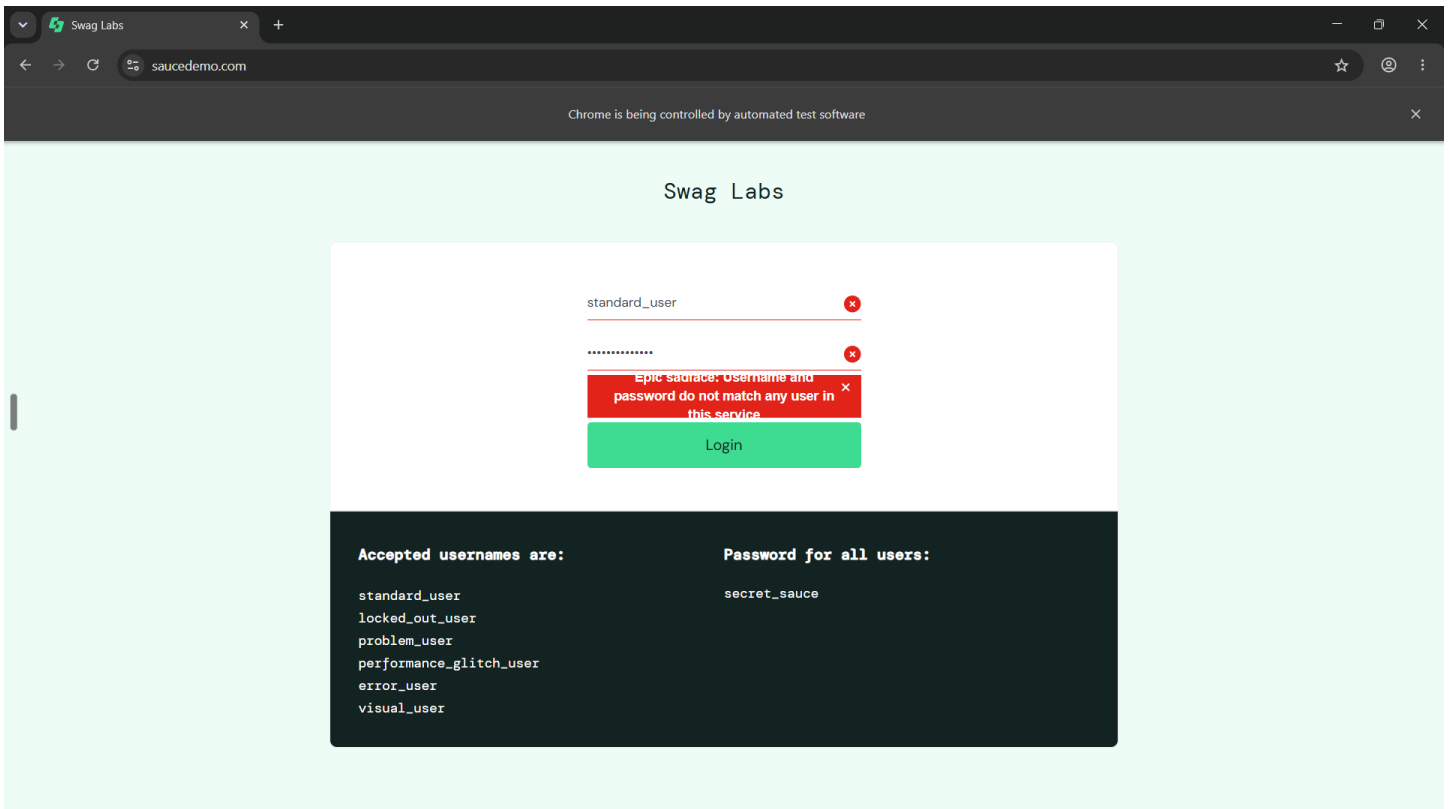
Python / Selenium Implementation

```
Q8.py X
Module 2 > Q8.py > ...
1  import csv
2  import time
3  from selenium import webdriver
4  from selenium.webdriver.common.by import By
5
6  driver = webdriver.Chrome()
7  driver.maximize_window()
8  print("Chrome should be open now")
9
10 with open("test_data.csv") as f:
11     rows = list(csv.DictReader(f))
12
13 for row in rows:
14     driver.get("https://www.saucedemo.com/")
15     driver.find_element(By.ID, "user-name").send_keys(row["username"])
16     driver.find_element(By.ID, "password").send_keys(row["password"])
17     driver.find_element(By.ID, "login-button").click()
18     time.sleep(3)
19     error = driver.find_element(By.CSS_SELECTOR, "h3[data-test='error']").text
20     print(row["username"], "/", row["password"], "->", error)
21     assert row["expected_error"] in error
22
23 print("ASSIGNMENT 8 - DDT LOGIN TEST PASSED")
24 time.sleep(3)
25 driver.quit()
26
```

Execution Evidence

```
Problems Output Debug Console Terminal Ports
• (venv) PS D:\PythonAutomation\Assignments\Module 2> python .\Q8.py
Chrome should be open now
locked_out_user / secret_sauce -> Epic sadface: Sorry, this user has been locked out.
standard_user / wrong_password -> Epic sadface: Username and password do not match any user in this service
/ secret_sauce -> Epic sadface: Username is required
standard_user / -> Epic sadface: Password is required
ASSIGNMENT 8 - DDT LOGIN TEST PASSED
❖ (venv) PS D:\PythonAutomation\Assignments\Module 2> |
```

Terminal output



Browser session showing one iteration of the data-driven loop (missing-password case).

ASSIGNMENT 9

PyTest Integration with HTML Reporting

Description

Convert the login checks into PyTest test functions so they can be discovered and executed by the PyTest runner. Run the test suite from the terminal using the pytest-html plugin to auto-generate a self-contained HTML execution report summarizing pass/fail results for each test.

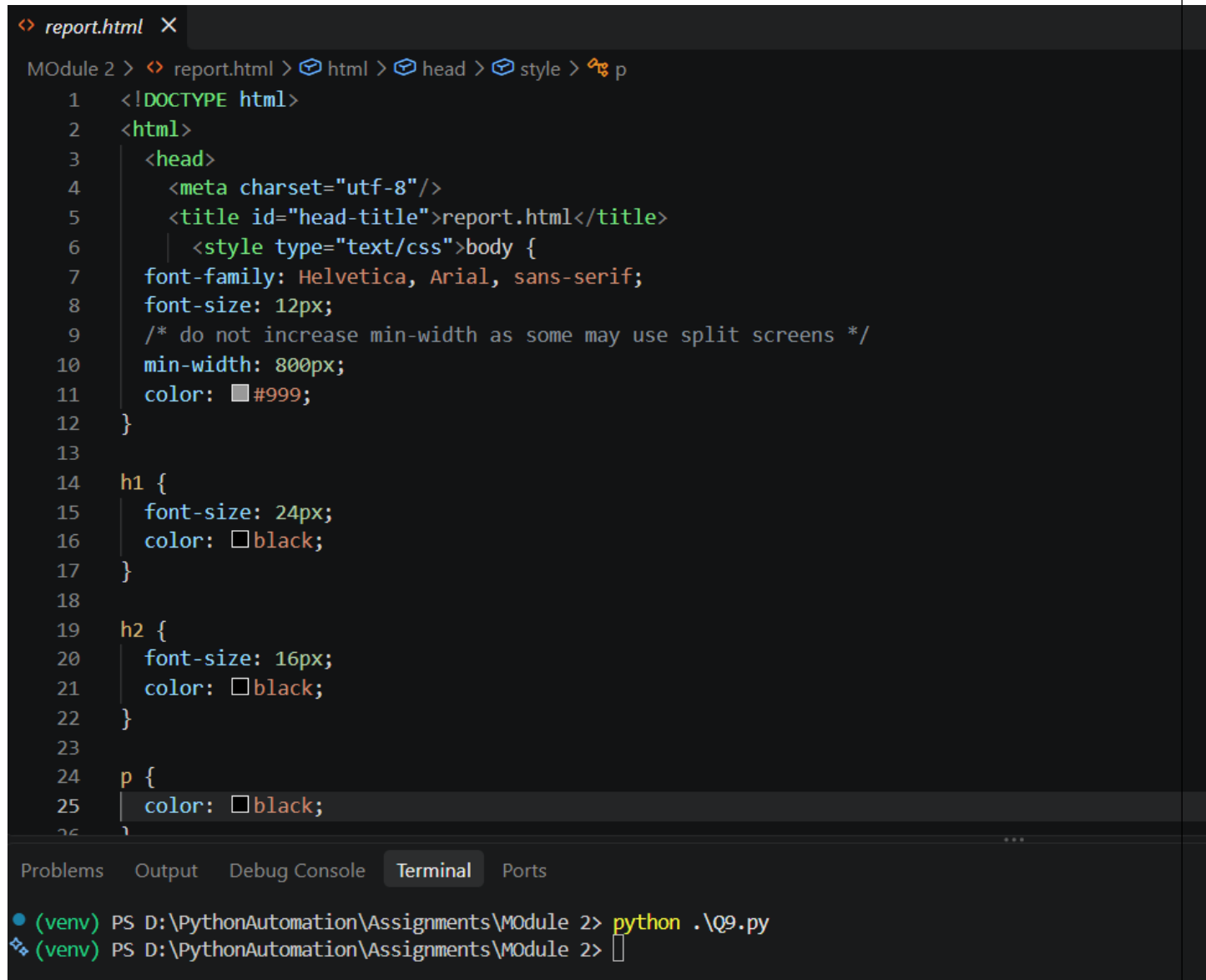
Python / PyTest Implementation (Q9.py)

```
Q9.py X
Module 2 > Q9.py > test_invalid_login
1  from selenium import webdriver
2  from selenium.webdriver.common.by import By
3
4  def test_valid_login():
5      driver = webdriver.Chrome()
6      driver.maximize_window()
7      driver.get("https://www.saucedemo.com/")
8      driver.find_element(By.ID, "user-name").send_keys("standard_user")
9      driver.find_element(By.ID, "password").send_keys("secret_sauce")
10     driver.find_element(By.ID, "login-button").click()
11     assert driver.current_url == "https://www.saucedemo.com/inventory.html"
12     driver.quit()
13
14     def test_invalid_login():
15         driver = webdriver.Chrome()
16         driver.maximize_window()
17         driver.get("https://www.saucedemo.com/")
18         driver.find_element(By.ID, "user-name").send_keys("locked_out_user")
19         driver.find_element(By.ID, "password").send_keys("secret_sauce")
20         driver.find_element(By.ID, "login-button").click()
21         error = driver.find_element(By.CSS_SELECTOR, "h3[data-test='error']").text
22         assert "locked out" in error
23         driver.quit()
```

Run Command

```
pytest -s Q9.py --html=report.html --self-contained-html
```

Execution Evidence



The image shows a VS Code editor window with a file named `report.html` open. The breadcrumb navigation at the top indicates the file path: `Module 2 > report.html > html > head > style > p`. The editor displays the following HTML code:

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8"/>
5     <title id="head-title">report.html</title>
6     <style type="text/css">body {
7       font-family: Helvetica, Arial, sans-serif;
8       font-size: 12px;
9       /* do not increase min-width as some may use split screens */
10      min-width: 800px;
11      color: #999;
12    }
13
14    h1 {
15      font-size: 24px;
16      color: black;
17    }
18
19    h2 {
20      font-size: 16px;
21      color: black;
22    }
23
24    p {
25      color: black;
```

Below the editor, the `Terminal` tab is active, showing the following commands and output:

```
(venv) PS D:\PythonAutomation\Assignments\Module 2> python .\Q9.py
(venv) PS D:\PythonAutomation\Assignments\Module 2>
```

